EDA using Power BI
Customer Churn Analysis Report

Name : Shamyuktha U T
Reg no : 727823TUAD141

1. Problem Statement

A telecom company providing subscription-based services has recently observed a decline in active customers, directly impacting recurring monthly revenue. Management suspects that customer churn is influenced by contract type, tenure duration, and monthly billing charges, but lacks clear analytical insights.

The organization requires a data-driven business intelligence solution to analyze customer behavior, identify churn patterns, determine major churn drivers, and support proactive retention strategies through visual analytics and actionable insights.

2. Project Objectives

- Analyze customer churn behavior using business intelligence techniques.
- Understand churn distribution across tenure groups and contract types.
- Identify financial and behavioral factors influencing churn.
- Develop interactive dashboards using Power BI.
- Calculate key performance indicators such as churn rate.
- Enable dynamic analysis using slicers (gender and contract type).
- Provide strategic recommendations to reduce customer churn.

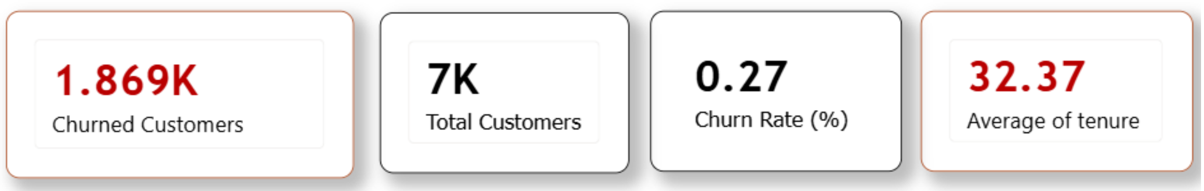
3. Tools & Technologies Used

- **Power BI** – Dashboard design and data visualization.
- **DAX (Data Analysis Expressions)** – KPI calculations such as churn rate.
- **Data Modeling** – Relationships and calculated fields.
- **Business Intelligence Techniques** – Pattern recognition and segmentation analysis.
- **Data Sources** – Telecom customer dataset (CSV/Excel).

4. Data Preparation & Modelling

- Cleaned and standardized customer data.
- Created calculated tenure groups:
  - 0–12 months
  - 13–24 months
  - 25–48 months
  - 49+ months
- Generated DAX measures for:
  - Churn rate
  - Customer counts
  - Average tenure
- Designed interactive slicers for:
  - Gender
  - Contract type
- Built relationships between churn status, tenure, contracts, and billing attributes.
- Structured the dashboard to highlight churn KPIs and patterns visually.

5. Executive Summary (KPIs)



Based on the Power BI dashboard:

- **Total Customers:** ~7K
- **Churned Customers:** ~1.869K
- **Overall Churn Rate:** ~27%
- **Average Customer Tenure:** ~32 months

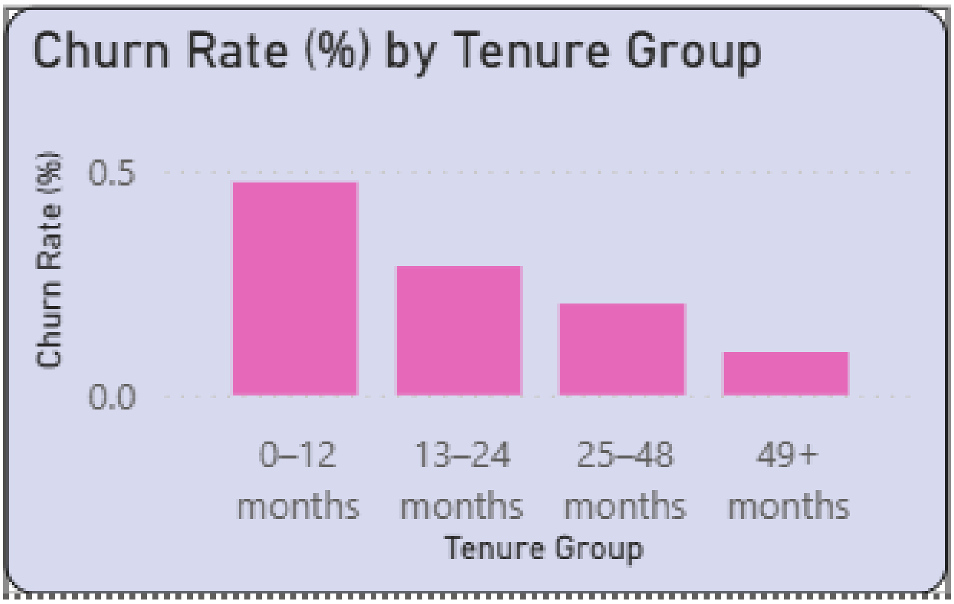
- **Current Churn Rate vs Target:** Below expected churn goal

Key Insights:

- A significant portion of customers churn early in their lifecycle.
- Contract flexibility strongly affects churn risk.
- Customers paying higher monthly charges show higher churn probability.

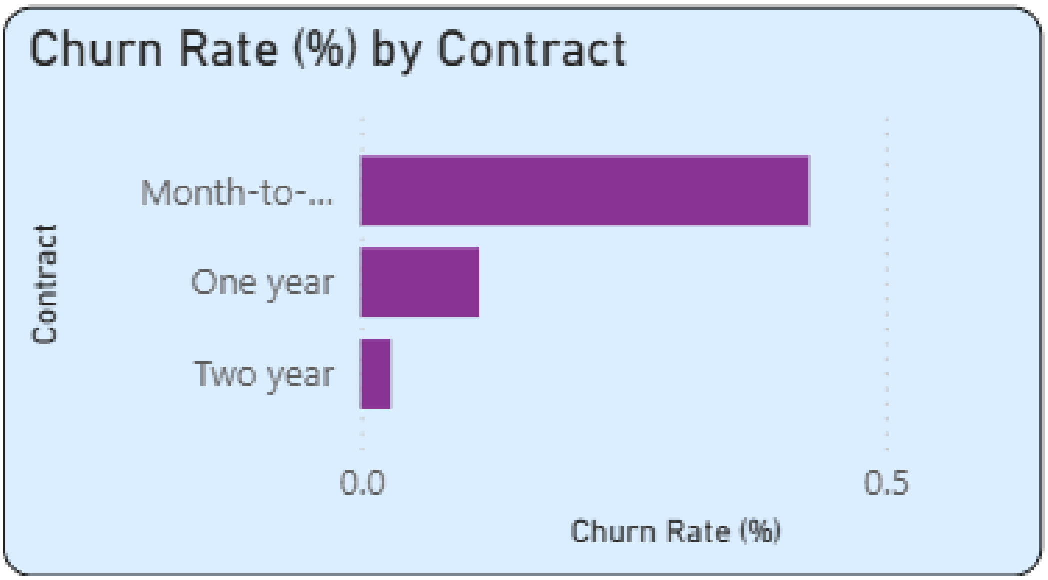
6. Analysis: Customer Tenure & Contract Behavior

Tenure Distribution Trends



- Customers in the **0–12 months group** have the highest churn rates.
- Churn steadily decreases as tenure increases.
- Customers with tenure above 49 months show strong loyalty.
- Early customer lifecycle is the most critical retention phase.

Contract Type Analysis



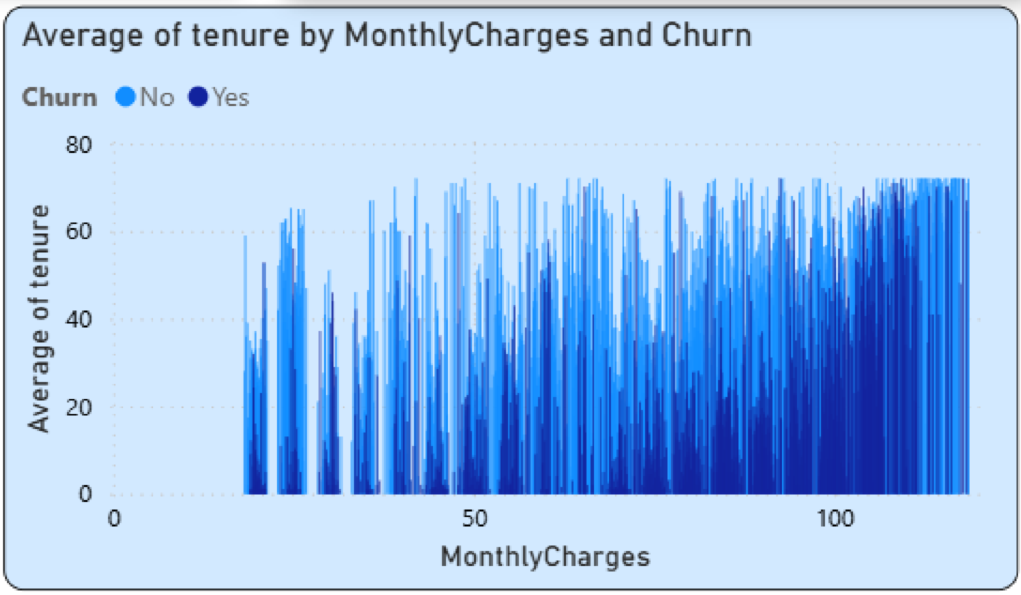
- **Month-to-month contracts** demonstrate the highest churn levels.
- One-year contracts show moderate churn behavior.
- Two-year contracts show the lowest churn risk.

Observation:

Long-term contracts increase customer commitment and reduce churn significantly.

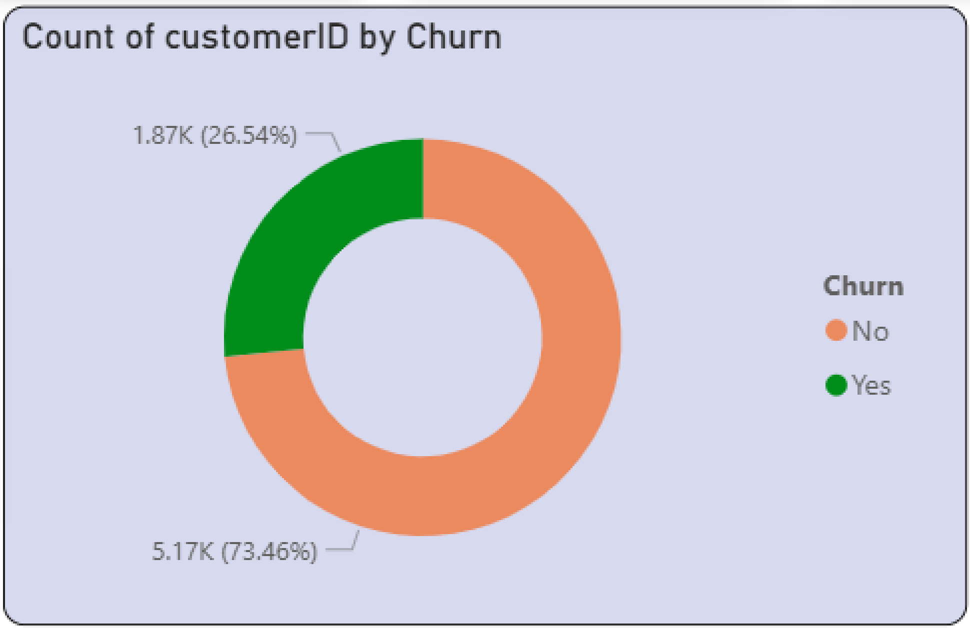
7. Financial & Behavioral Impact Analysis: Monthly Charges & Customer Patterns

Monthly Charges vs Churn



- Customers with higher monthly charges display increased churn rates.
- High-cost plans create price sensitivity and dissatisfaction risk.

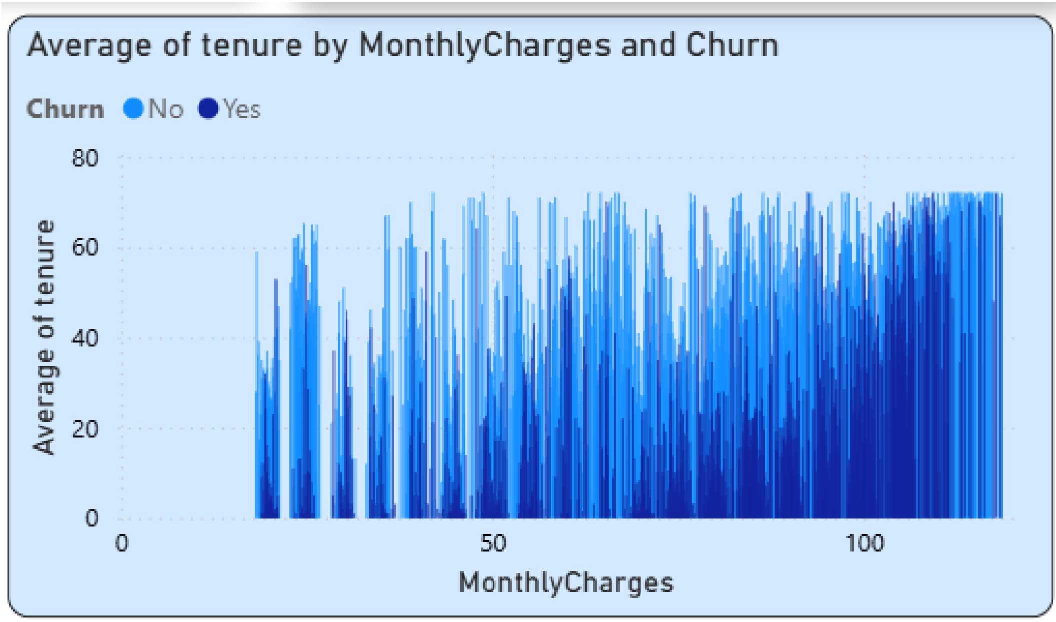
Behavioral Patterns



- Customers paying premium charges with short tenure churn faster.

- Flexible contracts combined with high charges increase churn probability.
- Billing-related customer experience may influence retention.

Revenue Risk Insight



- Early churn reduces lifetime value.
- High-paying customers leaving early impacts recurring revenue streams.

8. Strategic Recommendations

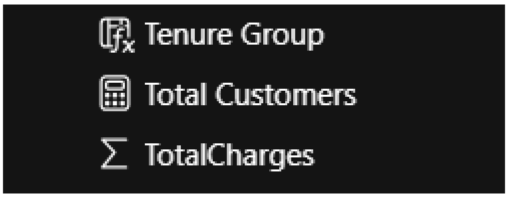
- Promote long-term contracts using discounts and loyalty benefits.
- Implement early-stage customer engagement programs during the first 12 months.
- Offer personalized pricing options for high-charge customers.
- Monitor high-risk churn segments using real-time dashboard insights.
- Introduce proactive retention campaigns targeting month-to-month customers.
- Improve onboarding experience to increase early customer satisfaction.

Steps Performed in Power BI

Step 1 – Data Loading

Action Performed:

- Imported telecom customer dataset into Power BI.
- Verified column types (text, numeric, categorical).



Caption:

Dataset loaded into Power BI showing customer attributes including tenure, contract type, churn status, and monthly charges.

Step 2 – Data Preparation

Action Performed:

- Checked for missing values.
- Created Tenure Groups:
  - 0–12 months
  - 13–24 months
  - 25–48 months
  - 49+ months
- Verified data consistency.

Tenure Group
0–12 months
0–12 months
0–12 months
0–12 months
0–12 months
0–12 months
0–12 months
0–12 months
0–12 months
0–12 months

```
Tenure Group =
SWITCH(
    TRUE(),
    Telco[tenure] <= 12, "0-12 months",
    Telco[tenure] <= 24, "13-24 months",
    Telco[tenure] <= 48, "25-48 months",
    "49+ months"
)

```

Caption:

Data preparation including calculated tenure grouping for churn analysis.

Step 3 – Creating DAX Measures

Churn Rate Measure

```
Churn Rate (%) =
DIVIDE(
   CALCULATE(COUNT('Telco'[customerID]), 'Telco'[Churn] = "Yes"),
   COUNT('Telco'[customerID])
```

)

```
Churn Rate (%) =
DIVIDE([Churned Customers], [Total Customers], 0)
```

Churned Customers Measure

Churned Customers =

CALCULATE(COUNT('Telco'[customerID]), 'Telco'[Churn] = "Yes")

```
Churned Customers =
CALCULATE(
    COUNT(Telco[customerID]),
    Telco[Churn] = "Yes"
)
```

Caption:

DAX measures created for churn analysis KPIs.

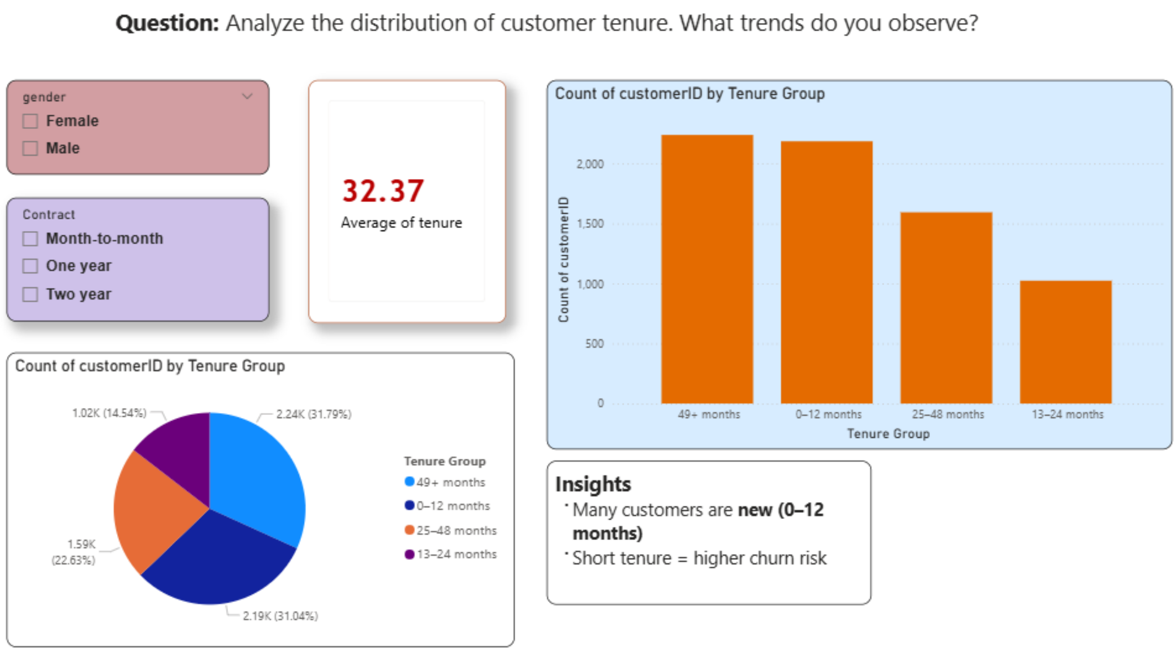
Step 4 – Dashboard Development

Created visuals:

- KPI Cards
- Donut Chart
- Bar Charts
- Monthly Charges Analysis Chart
- Slicers (Gender & Contract)

Question-Based Analysis & Answers

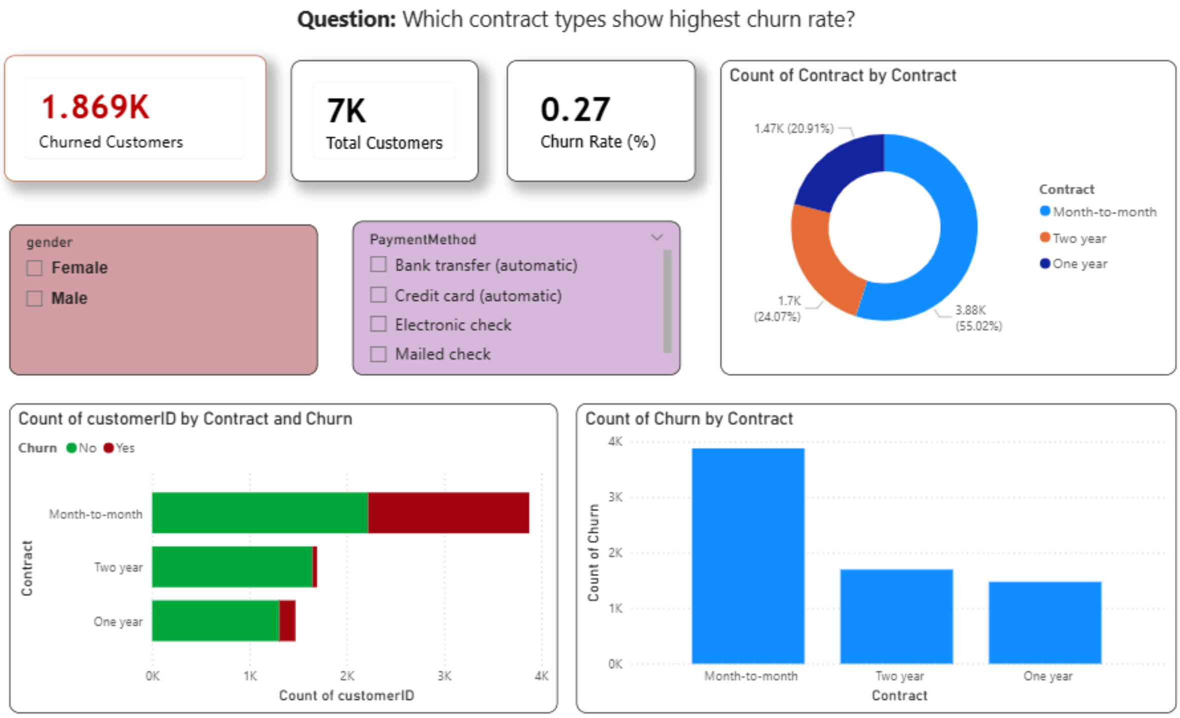
Q1 – Analyze the distribution of customer tenure. What trends do you observe?



Answer:

- Customers with tenure between 0–12 months show the highest churn rates.
- Churn decreases steadily as tenure increases.
- Customers with tenure above 49 months demonstrate strong loyalty.
- Early customer lifecycle is the most critical stage for retention strategies.

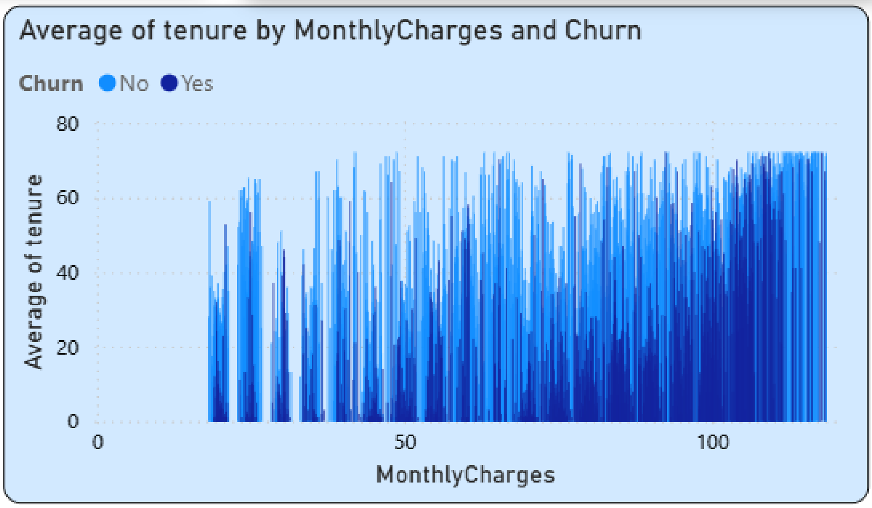
Q2 – Which contract types show the highest churn rate?



Answer:

- Month-to-month contracts show the highest churn rate.
- One-year contracts have moderate churn levels.
- Two-year contracts have the lowest churn rate.
- Long-term contracts significantly improve customer retention.

Q3 – Are monthly charges related to customer churn?



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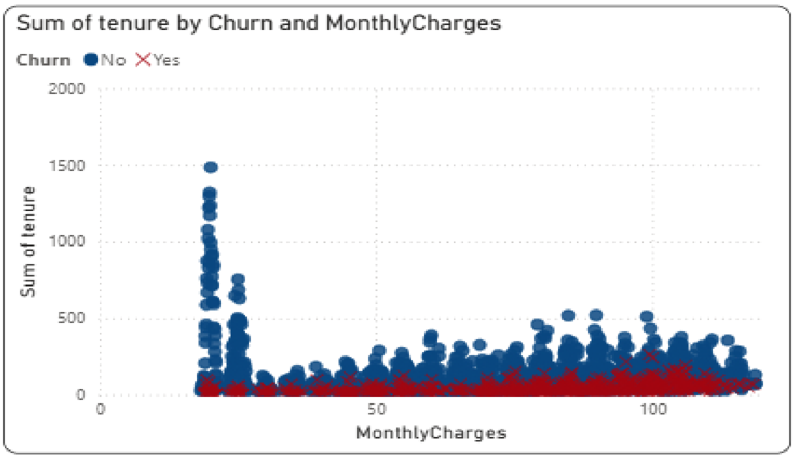


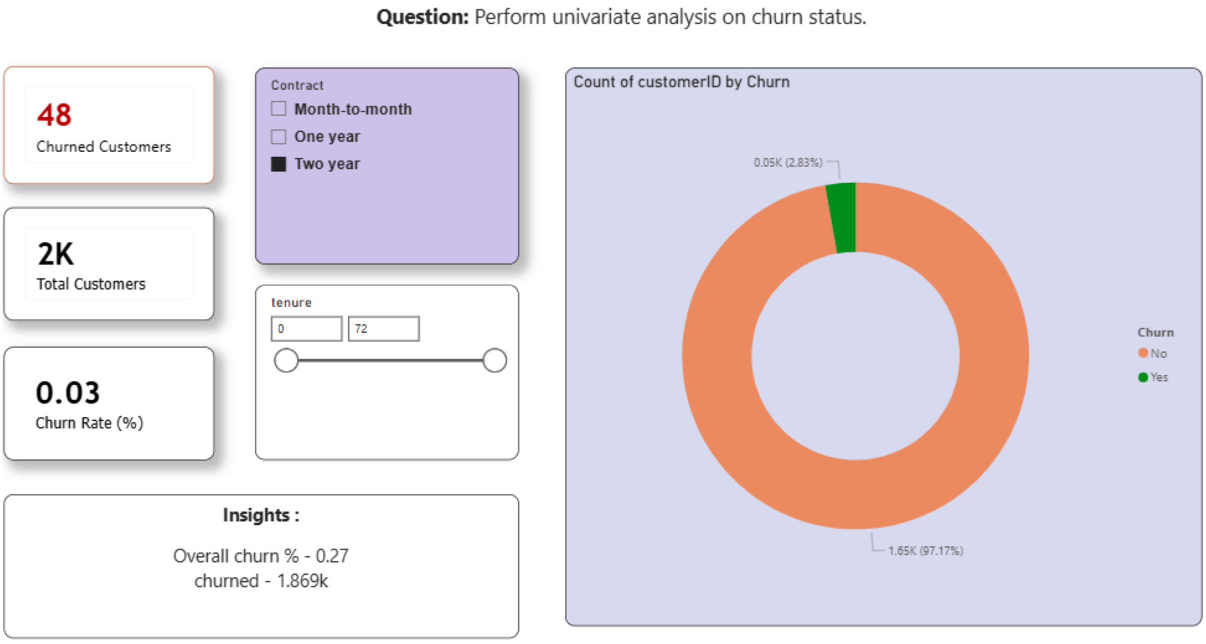
Insights

- Higher charges → higher churn
- Price-sensitive customers

Answer:

- Customers with higher monthly charges are more likely to churn.
- High charges combined with short tenure increase churn risk.
- Pricing sensitivity plays a major role in customer retention.

Q4 – Perform univariate analysis on churn status.



Answer:

- Approximately 27% of customers have churned.
- Around 73% remain active.
- This indicates a significant churn level requiring strategic action.

Q5 – Create a DAX measure to calculate Churn Rate.

```
Churn Rate (%) =
DIVIDE(
    CALCULATE(COUNT('Telco'[customerID]), 'Telco'[Churn] = "Yes"),
    COUNT('Telco'[customerID])
)
```

Explanation:

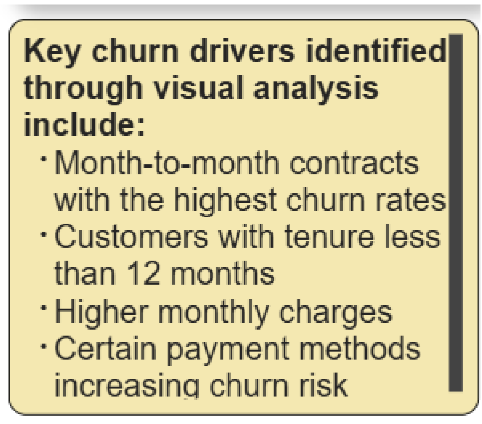
- Calculates percentage of churned customers over total customers.

Q6 – Use slicers to analyze churn by gender and contract type.

Answer:

- Gender slicer allows comparison of churn between male and female customers.
- Contract slicer helps identify churn risk based on contract duration.
- Interactive filtering supports targeted analysis.

Q7 – Identify key churn drivers using visualization patterns.



Answer:

- Month-to-month contracts.
- Customers with tenure less than 12 months.
- Higher monthly charges.
- Certain billing/payment behaviors.

Q8 – Which customer segment requires immediate retention strategies?

Answer:

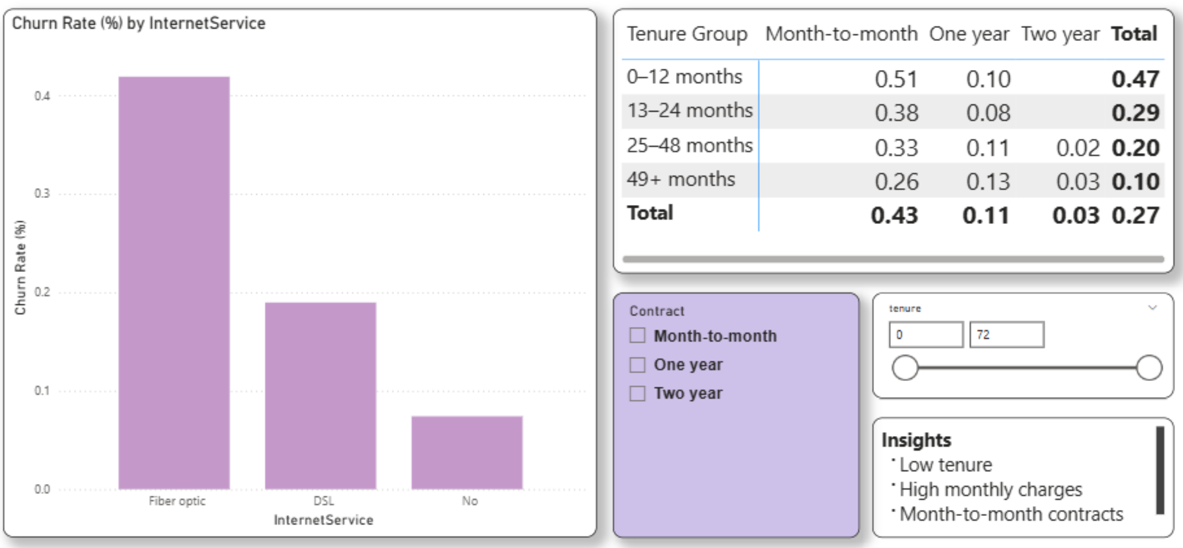
- Customers with:
  - Month-to-month contracts
  - Tenure below 12 months
  - High monthly charges
- These customers represent the highest churn risk segment.

Q9 – How can churn insights influence business decisions?

Answer:

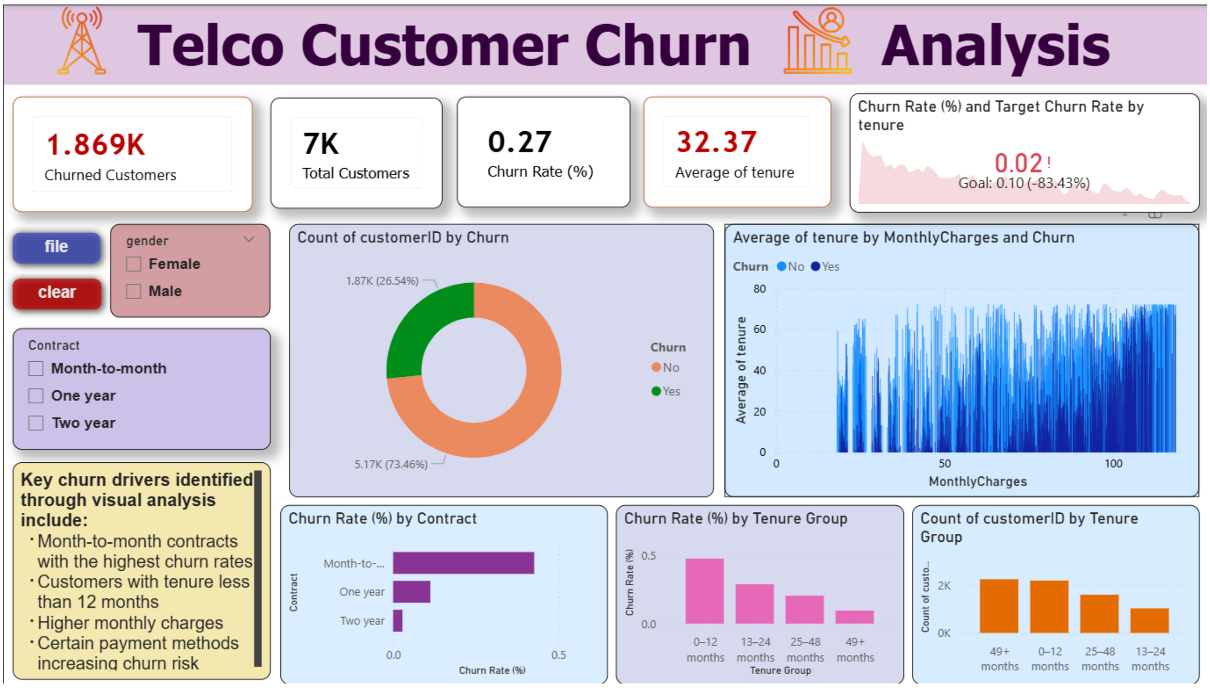
- Encourage long-term contracts with incentives.
- Implement early onboarding programs.
- Offer personalized pricing plans.
- Monitor churn risk using real-time dashboards.
- Improve customer engagement strategies.

Question: Identify key churn drivers using visualization patterns.



Tenure Group	Month-to-month	One year	Two year	Total
0–12 months	0.51	0.10		**0.47**
13–24 months	0.38	0.08		**0.29**
25–48 months	0.33	0.11	0.02	**0.20**
49+ months	0.26	0.13	0.03	**0.10**
Total	**0.43**	**0.11**	**0.03**	**0.27**

9. Conclusion



The Telco Customer Churn Analysis project successfully identifies major churn drivers through interactive Power BI dashboards. The analysis shows that short tenure periods, flexible contracts, and higher monthly charges significantly contribute to customer attrition.

By leveraging visual analytics and KPI-driven insights, the organization can proactively identify at-risk customers and implement targeted retention strategies. Applying these insights will improve customer loyalty, reduce churn rates, and strengthen long-term revenue sustainability.